

Mitigating Interpersonal Variability in Machine Learning Models for Human Behavior Monitoring

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Abstract

Machine learning models for human behavior monitoring typically rely on population-level data, assuming that physiological patterns generalize well across individuals. However, this assumption often fails due to interpersonal variability — differences in individuals' physiological responses and subjective interpretations influenced by personal, cultural, and psychological factors. This research first investigates how such variability impacts model performance in two scenarios: sleep monitoring, where label variability arises because individuals assign different subjective ratings to similar physiological patterns, and transport mode recognition, where signal variability occurs as individuals exhibit distinct sensor signals for the same activity. Our empirical findings show that population models struggle to generalize, especially in tasks involving subjective assessments. To overcome these challenges, we firstly propose generative modeling techniques to augment limited user-specific data, and secondly introduce a clustering-based strategy that enables scalable personalization by grouping users according to both physiological signal patterns and labeling behavior.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing theory, concepts and paradigms; • **Computing methodologies** → Machine learning.

Keywords

Interpersonal Variability, Human Behavior Modeling, Personalization, Generative Modeling, Clustering, Sleep, Transport Mode Recognition, Label Variability, Signal Variability

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1 Problem statement

Advances in sensing technologies and data analytics have significantly enhanced the ability to monitor and understand human behavior across various domains, notably healthcare, well-being,

and daily activity tracking. These devices continuously collect diverse physiological signals, such as heart rate, skin temperature, electrodermal activity, and movement patterns, offering significant potential to infer and understand human behaviors and states (e.g., stress, sleep quality, and transportation mode) [2, 4, 9]. Typically, machine learning (ML) algorithms leverage these data streams to recognize and predict human behaviors by learning patterns from datasets collected across broad populations. Such algorithms are referred to as *population models*, designed with the assumption that common patterns in physiological signals generalize effectively across individuals.

However, a critical challenge arises from the inherent differences between individuals in terms of their physiological responses and subjective perceptions, commonly referred to as *interpersonal variability*. This variability is influenced by factors such as personality traits [17], cultural differences [24], personal habits [3], and emotional responses [13]. Interpersonal variability manifests itself in both the physiological signals recorded by sensors [26], which we refer to as *signal variability*, and in the subjective evaluation of the task being performed [16], which we refer to as *label variability*. For example, for a system trained to predict stress levels from physiological data, signal variability occurs when the same stress label is associated with physiological data traces with very different characteristics, whereas label variability occurs when comparable physiological responses are associated with differing subjective labels.

The consequences of interpersonal variability are significant and pose major challenges for the effectiveness of population-based ML models. Models trained on data from a diverse population often underperform when applied to new individuals whose physiological characteristics or subjective labeling patterns differ substantially from those represented in the training dataset [18, 21]. One promising solution is the adoption of personalized models, which are explicitly tailored to individual users' physiological signals and subjective perceptions. Personalized models have demonstrated improved performance in various applications, such as sleep quality recognition, stress detection, and gait analysis [9, 18, 21]. Nevertheless, these personalized models typically face significant practical limitations, notably data scarcity, as acquiring and labeling sufficient user-specific data is costly, time-consuming, and challenging to maintain over extended periods. Additionally, many existing personalized approaches fail to simultaneously address both signal and label variability, which limits their effectiveness and adaptability in real-world applications.

Motivated by these critical limitations, my doctoral research systematically investigates the impact of interpersonal variability on ML models designed for human behavior monitoring, aiming



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to enhance their effectiveness and applicability. Initially, the research focuses specifically on two application scenarios — sleep monitoring, characterized by label variability due to subjective perceptions, and transport mode recognition, which primarily exhibits signal variability due to differences in movement patterns among individuals. Through the analysis of these scenarios, the research aims to deepen the understanding of how interpersonal variability affects ML model performance and to develop novel strategies that effectively mitigate these challenges.

The current and final phase of my research has two main objectives: to explore generative modeling techniques as a means of addressing the data scarcity challenge inherent in personalized modeling, and to develop clustering-based approaches that group individuals according to both physiological and subjective similarities. This strategy enables the creation of personalized yet scalable solutions that are robust to variability in both sensor signals and subjective labels.

2 Related Work

While interpersonal variability is often viewed as meaningful signal in psychology, in many machine learning pipelines, such variability is instead treated as *noise* — i.e., undesired fluctuations that obscure the true underlying relationship between input and output. Depending on where it is observed, it was typically categorized as either *label noise*, when subjective annotations exhibited variability [8], or as *feature noise*, when the input signals themselves exhibited undesirable variation [27]. This led to the widespread use of preprocessing techniques such as normalization, smoothing, or denoising to suppress individual-specific fluctuations in favor of more “generalizable” patterns. However, recent research has emphasized that such variability should not be dismissed as noise, but rather recognized as a meaningful and complementary layer of information, especially in domains involving subjective or contextual experiences [10, 14]. Eliminating variability does not resolve the underlying challenge of modeling interpersonal and intrapersonal differences — in fact, it often removes critical nuances that are essential for accurate and fair inference [25]. Consequently, there is a growing shift toward *personalized modeling* approaches, which treat individual variability as a signal to be leveraged rather than suppressed. Personalization allows models to adapt to each individual’s unique traits and dynamics.

Li et al. [11] proposed a comprehensive taxonomy of state-of-the-art personalization approaches, categorizing them into the following groups: user-specific modeling, group-specific modeling, weighting-based methods, fine-tuning strategies, multitask learning, feature augmentation, and generative modeling techniques.

User-specific modeling refers to the development of a dedicated model for each individual, trained exclusively on that user’s data. This approach is exemplified by works such as [20]. Collecting data to train a user-specific model requires time and resources from both the system, which gathers extensive data from each participant, and the user, who needs to provide numerous labels before gaining benefits from the system.

Group-specific modeling, such as Taylor et al. [23], involves clustering similar users together, reducing the number of models required. However, most existing techniques cluster users solely based

on signal similarity, without accounting for variability in subjective labels.

Weighting-based approaches concentrate on assigning weights to data points based on their similarity to a specific user. For instance, Chattopadhyay et al. [5] demonstrated how this approach helped detect early fatigue in adults. This technique performs effectively when there are negligible differences across users, as it involves using all instances during training, albeit with varying weights.

Fine-tuning techniques utilize a pretrained population model, which is adapted to meet the needs of a specific user. Rescigno et al. [15] showed how fine-tuning a population model improved valence/arousal detection accuracy. This method assumes that user-specific and pretrained population models share parameters or exhibit similar hyperparameter distributions, though this assumption does not always hold.

Multitask learning involves training a single model to handle multiple related tasks at the same time. This approach focuses on having multiple output layers, each corresponding to a different task. According to Li et al. [11], the most common types are *user-as-task* [22] and *cluster-as-task* [6]. For example, in user-as-task, each user is treated as a distinct task, and the model includes sharing layers to learn shared features and personalized layers to tailor inferences for each user. However, the effectiveness of this approach relies on the presence of common patterns between the multiple tasks that the model is solving; otherwise, shared learning can adversely affect performance.

Feature augmentation techniques involve adding additional features or information to enhance the performance of systems. These techniques often employ generative models to generate additional user-specific features. For instance, Salam et al. [19] first utilize a model for personality inference and subsequently perform engagement detection. These methods require prior knowledge to identify features that improve task performance, which is challenging in domains lacking a comprehensive understanding of the critical features necessary to enhance performance.

Generative-based methods follow a two-step process. First, a model is developed to generate new samples from the target user. Then, a second model is trained using a mixture of real and generated data. For example, Yang et al. [28] employed a generative adversarial network (GAN) to generate additional samples and a convolutional neural network (CNN) for facial expression recognition. Generative-based models leverage the fact that machine learning benefits from large and diverse training datasets. However, there is a risk that unrealistic signals — i.e., a signal that deviates from real-world patterns — may be generated, which could impact the model by introducing noise.

User- and group-specific modeling are trained on data from the target user or a subset of the most similar users. Consequently, they do not incorporate data from all other users during training. This selective approach exacerbates an already existing data scarcity issue due to the time and resources required for data collection. For instance, to develop a model to predict sleep quality at a population level using a five-class classification system based on a 5-point Likert scale [12] (ranging from 1 to 5), extensive data collection is required. Existing studies typically design data collection protocols to include at least several dozen labels per class to train a model that not only performs well on the users seen during training but also

generalizes to new users with similar characteristics. To obtain 20 labels per class (i.e., ratings of 1, 2, 3, 4, and 5) for each participant, data collection must span a minimum of $20 * 5 = 100$ days. However, this estimate often underrepresents the actual duration needed for two key reasons. First, extreme events — such as poor sleep quality — are infrequent for some individuals, making it difficult to collect balanced data across all classes. The second reason is that individuals might not comply with data collection protocols and even forget to provide data on a daily basis. To address the resulting data scarcity challenge, the first proposed solution involves the use of generative models based on style transfer or encoder-decoder architectures to create synthetic samples that closely resemble data from the target user.

In addition, while personalized modeling is often used to address sensor variability — typically by clustering users based on similarities in signal patterns or extracted features — our approach extends this idea further. We propose a clustering technique that accounts for both sensor and label variability by grouping users not only according to physiological signal similarity but also based on demographic attributes and the labeling patterns observed within each cluster.

3 Methodology

This research investigates the impact of interpersonal variability on ML models for human behavior recognition. We initially focused on two distinct application scenarios: sleep monitoring and transport mode detection. The methodological approach is divided into two parts: an initial *completed phase*, emphasizing a systematic empirical investigation of how interpersonal variability affects model performance, and a subsequent *proposed phase*, dedicated to the development and evaluation of novel methodological solutions addressing the identified challenges. Specifically, completed research activities have primarily examined the impact of interpersonal variability on existing machine learning approaches and identified two major issues in current solutions: (1) the data scarcity problem, which is exacerbated in user-specific modeling, and (2) the lack of approaches that account for both signal and label variability.

3.1 Completed Work: Investigating the Impact of Interpersonal Variability

Sleep Monitoring and Label Variability. Initially, we explored interpersonal variability in the context of sleep monitoring, specifically targeting the recognition of sleep quality and sleep/wake stages using physiological data collected from wearable devices (Empatica E4 [7]). For this purpose, we collected a dedicated dataset (*M2Sleep*), comprising 6557 hours of physiological data (accelerometer, electrodermal activity, skin temperature) and daily self-reported sleep quality ratings from 16 participants over one month. The dataset is available to the research community upon request and signature of a data sharing agreement, as detailed in [9]. As part of this work, I designed the data collection protocol, managed participant recruitment, and conducted the primary data analysis, which formed the core of my Master’s thesis. Using this dataset, we empirically compared personalized and population-based models to identify conditions under which personalization enhances inference performance. Our results demonstrated significant improvements

from personalized modeling specifically for subjective tasks, such as sleep quality recognition, highlighting the pronounced impact of label variability. However, limited benefits were observed for sleep/wake classification, an objective task primarily influenced by physiological features [9].

Objective and Subjective Sleep Quality Analysis. Building on the collected dataset, we have further examined label variability by analyzing relationships between subjective sleep quality ratings and objective physiological measures. This analysis revealed that self-reported sleep quality ratings correlated only weakly with conventional sensor-derived features (e.g., sleep duration, skin temperature, electrodermal activity metrics). Additionally, personality traits such as extroversion, agreeableness, and baseline sleep quality (e.g., PSQI scores) significantly influenced subjective sleep evaluations, emphasizing the necessity to explicitly incorporate subjective variability in sleep monitoring systems [1].

Transport Mode Recognition and Signal Variability. We next investigated interpersonal variability in transport mode classification, primarily driven by signal variability. Utilizing the Sussex-Huawei Locomotion-Transportation (SHL) dataset, we proposed an enhanced ML pipeline, *DecayXGBoost*, which integrated heuristic smoothing techniques into the standard XGBoost classifier. Although the proposed model exhibited robust overall performance, its accuracy declined when evaluated on unseen users. This performance drop highlighted substantial interpersonal variability in movement patterns, underscoring the challenges of generalizing models trained solely on population-level data [2].

3.2 Proposed Work: Developing Solutions to Interpersonal Variability

Based on the insights from the empirical investigations, the proposed phase of this research focuses on developing methods to mitigate the limitations caused by interpersonal variability — particularly in settings where data is scarce and user-specific models are impractical. The aim is to create scalable personalization strategies that can handle both signal and label variability in real-world applications. Two directions are being pursued:

Generative Modeling to Address Data Scarcity. A major barrier to personalized modeling is the limited amount of labeled data available per user. To alleviate this, we propose using generative modeling techniques to create synthetic physiological data that reflect user-specific patterns. Approaches such as style transfer and encoder-decoder architectures will be explored to simulate realistic signal traces, thereby augmenting training data for personalized models. The goal is to reduce the need for long data collection while still capturing the accuracy and individual differences needed for reliable behavior recognition.

Clustering Users Based on Physiological and Subjective Similarity. User-specific models can achieve high accuracy by effectively capturing individual variations in physiological responses and subjective labeling patterns. They are especially valuable when interpersonal variability is so distinct that data from others fails to generalize. However, these models are difficult to scale and still

require labeled data from each new user, making them impractical in many real-world scenarios. To address these limitations, we propose a clustering-based approach that enables group-level personalization by grouping users with similar physiological responses and subjective labeling behaviors. Unlike most existing methods that focus solely on signal features, our approach integrates both sensor-derived signals (e.g., movement, electrodermal activity) and self-reported attributes (e.g., personality traits, emotional states) to account for both signal and label variability.

4 Evaluation

The evaluation strategy assesses the impact of interpersonal variability on classification performance and tests the effectiveness of the proposed generative modeling and clustering approaches against current state-of-the-art solutions. We evaluate model performance using balanced accuracy and macro F1-score, employing leave-one-subject-out (LOSO) validation to measure generalization to unseen users, and few-shot learning protocols to assess robustness in data-scarce scenarios. To evaluate generative modeling, we compare models trained on real user data, real data augmented with synthetic samples, and synthetic data alone, in order to quantify the benefits of data augmentation. For clustering-based personalization, we benchmark population models, user-specific models, signal-based clusters, and our proposed hybrid clustering approach, which groups users based on both physiological signal similarity and subjective label patterns.

5 Contribution

This research advances both the understanding of interpersonal variability in human behavior modeling and the development of scalable and effective methods for personalization.

First, the empirical work conducted so far demonstrates that population-based models struggle to generalize across individuals, particularly in subjective tasks such as sleep quality recognition. These findings highlight the critical role of personalization and provide quantitative evidence of how different types of variability — signal and label — impact model performance.

Second, the research proposes novel methods to enable effective personalization with minimal reliance on user-specific data. These include a generative modeling approach to synthetically augment individual datasets, and a clustering-based framework that enables model sharing among users with similar physiological and subjective characteristics. Together, these approaches aim to reduce the data burden while maintaining high inference accuracy.

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References

- [1] Lidia Alecci, Nouran Abdalazim, Leonardo Alchieri, Shkurta Gashi, and Silvia Santini. 2022. On the Mismatch between Measured and Perceived Sleep Quality. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. ACM, Cambridge United Kingdom, 148–152. doi:10.1145/3544793.3563412
- [2] Lidia Alecci, Leonardo Alchieri, Nouran Abdalazim, Pietro Barbiero, Silvia Santini, and Martin Gjoreski. 2023. Enhancing XGBoost with Heuristic Smoothing for Transportation Mode and Activity Recognition. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing*. ACM, Cancun, Quintana Roo Mexico, 540–545. doi:10.1145/3594739.3610752
- [3] Lorenzo Alessio, Pietro Apostoli, and Michela Crippa. 1995. Influence of Individual Factors and Personal Habits on the Levels of Biological Indicators of Exposure. *Toxicology Letters* 77, 1 (May 1995), 93–103. doi:10.1016/0378-4274(95)03277-0
- [4] Yekta S. Can, Bert Arnrich, and Cem Ersoy. 2019. Stress Detection in Daily Life Scenarios Using Smart Phones and Wearable Sensors: A Survey. *Journal of Biomedical Informatics* 92 (April 2019), 103139. doi:10.1016/j.jbi.2019.103139
- [5] Rita Chattopadhyay, Qian Sun, Wei Fan, Ian Davidson, Sethuraman Panchanathan, and Jieping Ye. 2012. Multisource Domain Adaptation and Its Application to Early Detection of Fatigue. *ACM Transactions on Knowledge Discovery from Data* 6, 4 (Dec. 2012), 18:1–18:26. doi:10.1145/2382577.2382582
- [6] Vetha V. Chithra Raghuram, Hanan Salam, Jauwairia Nasir, Barbara Bruno, and Oya Celiktutan. 2022. Personalized Productive Engagement Recognition in Robot-Mediated Collaborative Learning. In *Proceedings of the 2022 International Conference on Multimodal Interaction (ICMI '22)*. Association for Computing Machinery, New York, NY, USA, 632–641. doi:10.1145/3536221.3556569
- [7] Empatica Inc. 2015. E4 Wearable Research Device. <https://www.empatica.com/en-eu/research/e4/>. Last accessed on November 25, 2024.
- [8] Benoit Frenay and Michel Verleysen. 2014. Classification in the Presence of Label Noise: A Survey. *IEEE Transactions on Neural Networks and Learning Systems* 25, 5 (May 2014), 845–869. doi:10.1109/TNNLS.2013.2292894
- [9] Shkurta Gashi, Lidia Alecci, Elena Di Lascio, Maïke E. Debus, Francesca Gasparini, and Silvia Santini. 2022. The Role of Model Personalization for Sleep Stage and Sleep Quality Recognition Using Wearables. *IEEE Pervasive Computing* 21, 2 (April 2022), 69–77. doi:10.1109/MPRV.2022.3164334
- [10] Salar Jahedi and Fabio Méndez. 2014. On the Advantages and Disadvantages of Subjective Measures. *Journal of Economic Behavior & Organization* 98 (Feb. 2014), 97–114. doi:10.1016/j.jebo.2013.12.016
- [11] Jialin Li, Alia Waleed, and Hanan Salam. 2023. A Survey on Personalized Affective Computing in Human-Machine Interaction. arXiv:2304.00377 [cs] doi:10.48550/arXiv.2304.00377
- [12] Rensis Likert. 1932. A Technique for the Measurement of Attitudes. *Archives of Psychology* 22 140 (1932), 55–55.
- [13] Rollin McCraty, Mike Atkinson, William A. Tiller, Glen Rein, and Alan D. Watkins. 1995. The Effects of Emotions on Short-Term Power Spectrum Analysis of Heart Rate Variability. *The American Journal of Cardiology* 76, 14 (Nov. 1995), 1089–1093. doi:10.1016/S0002-9149(99)80309-9
- [14] Waishan Qiu, Wenjing Li, Xun Liu, and Xiaokai Huang. 2022. Subjectively Measured Streetscape Qualities for Shanghai with Large-Scale Application of Computer Vision and Machine Learning. In *Proceedings of the 2021 Digital FUTURES*, Philip F. Yuan, Hua Chai, Chao Yan, and Neil Leach (Eds.). Springer, Singapore, 242–251. doi:10.1007/978-981-16-5983-6_23
- [15] Martina Rescigno, Matteo Spezialetti, and Silvia Rossi. 2020. Personalized Models for Facial Emotion Recognition through Transfer Learning. *Multimedia Tools and Applications* 79, 47 (Dec. 2020), 35811–35828. doi:10.1007/s11042-020-09405-4
- [16] Daniel Roggen, Kilian Forster, Alberto Calatroni, Andreas Bulling, and Gerhard Troster. 2010. On the Issue of Variability in Labels and Sensor Configurations in Activity Recognition Systems. In *Workshop at the 8th International Conference on Pervasive Computing (Pervasive 2010)*, 2010. Helsinki, 1–4.
- [17] Jennifer J. Russell, D. S. Moskowitz, David C. Zuroff, Debbie Sookman, and Joel Paris. 2007. Stability and Variability of Affective Experience and Interpersonal Behavior in Borderline Personality Disorder. *Journal of Abnormal Psychology* 116, 3 (Aug. 2007), 578–588. doi:10.1037/0021-843X.116.3.578
- [18] Aaqib Saeed, Tanir Ozcelebi, Johan Lukkien, Jan B.F. van Erp, and Stojan Trajanovski. 2018. Model Adaptation and Personalization for Physiological Stress Detection. In *2018 IEEE 5th International Conference on Data Science and Advanced Analytics (DSAA)*. IEEE, Turin, Italy, 209–216. doi:10.1109/DSAA.2018.00031
- [19] Hanan Salam, Oya Çeliktutan, Isabelle Hupont, Hatice Gunes, and Mohamed Chetouani. 2017. Fully Automatic Analysis of Engagement and Its Relationship to Personality in Human-Robot Interactions. *IEEE access : practical innovations, open solutions* 5 (2017), 705–721. doi:10.1109/ACCESS.2016.2614525
- [20] Rutvik V. Shah, Gillian Grennan, Mariam Zafar-Khan, Fahad Alim, Sujit Dey, Dhakshin Ramanathan, and Jyoti Mishra. 2021. Personalized Machine Learning of Depressed Mood Using Wearables. *Translational Psychiatry* 11, 1 (June 2021), 1–18. doi:10.1038/s41398-021-01445-0

- [21] Abolfazl Soltani, Hooman Dejnabadi, Martin Savary, and Kamiar Aminian. 2020. Real-World Gait Speed Estimation Using Wrist Sensor: A Personalized Approach. *IEEE journal of biomedical and health informatics* 24, 3 (March 2020), 658–668. doi:10.1109/JBHI.2019.2914940
- [22] Sara Taylor, Natasha Jaques, Ehimwenma Nosakhare, Akane Sano, and Rosalind Picard. 2020. Personalized Multitask Learning for Predicting Tomorrow’s Mood, Stress, and Health. *IEEE transactions on affective computing* 11, 2 (2020), 200–213. doi:10.1109/TAFFC.2017.2784832
- [23] Sandra L. Taylor, MaryBeth Lawless, Terese Curri, Soman Sen, David G. Greenhalgh, and Tina L. Palmieri. 2014. Predicting Mortality from Burn Injuries: The Need for Age-Group Specific Models. *Burns : journal of the International Society for Burn Injuries* 40, 6 (Sept. 2014), 1106–1115. doi:10.1016/j.burns.2014.03.010
- [24] Rachel Sing-Kiat Ting, Yue-Yun Aw Yong, Min-Min Tan, and Chee-Khong Yap. 2021. Cultural Responses to Covid-19 Pandemic: Religions, Illness Perception, and Perceived Stress. *Frontiers in Psychology* 12 (2021).
- [25] Arabella C. Vaughan and Damian P. Birney. 2023. Within-Individual Variation in Cognitive Performance Is Not Noise: Why and How Cognitive Assessments Should Examine Within-Person Performance. *Journal of Intelligence* 11, 6 (June 2023), 110. doi:10.3390/jintelligence11060110
- [26] Weichen Wang, Gabriella M. Harari, Rui Wang, Sandrine R. Müller, Shayan Mirjafari, Kizito Masaba, and Andrew T. Campbell. 2018. Sensing Behavioral Change over Time: Using Within-Person Variability Features from Mobile Sensing to Predict Personality Traits. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 2, 3 (Sept. 2018), 141:1–141:21. doi:10.1145/3264951
- [27] H. Xiong, Gaurav Pandey, M. Steinbach, and Vipin Kumar. 2006. Enhancing Data Analysis with Noise Removal. *IEEE Transactions on Knowledge and Data Engineering* 18, 3 (March 2006), 304–319. doi:10.1109/TKDE.2006.46
- [28] Huiyuan Yang, Zheng Zhang, and Lijun Yin. 2018. Identity-Adaptive Facial Expression Recognition through Expression Regeneration Using Conditional Generative Adversarial Networks. In *2018 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018)*. IEEE, Xi’an, China, 294–301. doi:10.1109/FG.2018.00050